

PrimePy: Number Theory Library

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Contents

1	Introduction	2
1.1	Philosophy	2
2	Algebra	2

1 Introduction

PrimePy is a Python number theory library with C++ backend, designed for high-performance computation and symbolic experimentation.

It is structured into three primary components:

1. Analytic Number Theory
2. Algebraic Number Theory
3. Cryptographic Algorithms

The PrimePy project is guided by the following Philosophy:

- **Pythonic interface:** A simple and flexible Python API that feels native and encourages experimentation.
- **High-performance C++ backend:** Core computations are implemented in C++ for speed and scalability.
- **Seamless visualization:** Built-in support for intuitive and customizable visualization of number-theoretic objects and results.
- **Minimal and general abstractions:** Mathematical structures are defined in a general and reusable way, without unnecessary overhead.

1.1 Philosophy

PrimePy is designed with several principles in mind:

1. Data Types — Precision Over Approximation TODO

2. Parallel Computation — Scalar Logic, Vector Execution All scalar operations in the library are automatically extended to **vectorized operations**, allowing users to apply the same logic to entire collections of values with minimal effort.

For example, consider the method `.power(a, n)` which computes a^n :

- If `a` is a scalar and `n` is a scalar, it returns a^n .
- If `a` is a list $[a_1, a_2, \dots, a_k]$ and `n` is a scalar m , the function returns $[a_1^m, a_2^m, \dots, a_k^m]$.
- If both `a` and `n` are lists of the same length, the function performs element-wise exponentiation:

$$\text{power}([a_1, \dots, a_k], [n_1, \dots, n_k]) \rightarrow [a_1^{n_1}, \dots, a_k^{n_k}]$$

2 Algebra

The library uses several